Simulating Trust Dynamics Under Delayed Governance in AI-Driven Platforms

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KEYWORDS

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ABSTRACT

As AI systems move into areas that have real social consequences, trust stops being something you can attribute to an algorithm in isolation. It starts to look more like an outcome of how the whole system behaves over time, including how feedback loops form, how quickly responses arrive, and how institutions react when things go wrong. In this work, we propose a simulation-based approach to study how trust evolves in AI-enabled socio-technical systems. We use Markovian Agent Models (MAMs) to represent users, recommender algorithms, and oversight bodies as interacting populations that change continuously over time.

We apply this framework to a misinformation scenario in online recommender systems, with particular attention to situations where engagement-driven optimization allows harmful content to spread before corrective actions take effect. Trust is modeled explicitly as an evaluative index derived from the system's harmful and corrective states, not as an abstract notion. The simulations suggest that trust erosion is driven mainly by slow or delayed institutional responses, more than by the behavior of individual algorithms on their own. Experiments that remove or delay parts of the governance machinery show that moderation and policy enforcement cannot replace one another: disabling either leads to long-lasting trust loss, and even moderate delays fail to change the system's trajectory once harmful dynamics have taken hold. The broader point is that trust is shaped by how feedback and timing interact at the system level, and that formal simulation can make these dependencies visible when testing different governance choices under clearly stated assumptions.

1 Introduction

Artificial Intelligence (AI) systems increasingly shape socially consequential processes, especially in the selection, ranking, and distribution of online content. In these settings, trust cannot be reduced to accuracy or short-term performance alone. It depends on how algorithmic choices interact over time with user behavior and institutional oversight, how quickly harmful dynamics are detected, and whether corrective action arrives before those dynamics become entrenched.

This paper approaches trust as a system-level outcome of interacting socio-technical processes. We model users, recommendation algorithms, moderation mechanisms, and platform-level policy enforcement as interacting populations evolving in continuous time. To do so, we adopt Markovian Agent Models (MAMs, Bobbio et al. (2016)), which provide a compact formalism for representing state-dependent transitions, delayed interventions, and feedback coupling across heterogeneous agent classes. Within this framework, trust does not influence behavior directly; instead, it is used as an evaluative construct for comparing system trajectories under an explicit notion of collective, institutional trust.

We instantiate the framework through a case study on misinformation propagation in online recommender systems. The focus is on a deliberately conservative stress-test scenario in which engagement-driven optimization amplifies harmful content before moderation and policy responses can intervene. The aim is to isolate unfavorable conditions under which governance mechanisms are placed under strain by delayed response. The results should therefore be read as claims about this stress-test regime, rather than as a general description of trust dynamics across all AI-driven platforms.

For this setting, we derive a closed system of differential equations and define a trust index from the system's harmful and corrective states. Under this operationalization, the simulations indicate that degradation of the evaluated trust index is driven mainly by delayed institutional response, not by algorithmic configuration in isolation. Ablation experiments further show that moderation and policy enforcement play complementary roles: removing either produces persistent trust degradation, whereas their joint presence allows partial recovery. These results highlight the importance of timing and coordination across technical and institutional layers in shaping trust-related outcomes.

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The notion of trust used in this paper is explicitly operational and model-relative. We do not treat trust as an individual psychological attitude, nor do we claim to instantiate a validated organizational-science measure. Instead, trust is used as a system-level evaluative observable that summarizes whether harmful dynamics or corrective institutional responses dominate along a trajectory. Its purpose is comparative: it allows governance configurations to be examined under explicit assumptions within a common formal framework.

The contribution of the paper is therefore to provide a formal framework in which trust-related outcomes can be evaluated at the level of the whole socio-technical system. Compared with prior simulation studies that examine recommendation effects, opinion dynamics, or misinformation diffusion separately, the present approach models users, recommender dynamics, moderation, and policy enforcement within a single continuous-time Markovian architecture. This makes it possible to represent delayed institutional response explicitly, separate the roles of different governance layers through ablation, and analyze how timing and coordination shape the resulting trust trajectory. In this sense, the paper contributes both a modeling perspective and a reproducible simulation framework for studying trust under delayed governance assumptions.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 introduces the model. Section 4 presents the simulation results. Section 5 discusses their implications, and Section 6 closes the paper.

2 Related Work

Simulation has been widely used to study how algorithmic systems shape collective behavior, especially in recommender platforms and misinformation diffusion. For example, Bellogín and Deldjoo (2021) analyze recommender systems operating on datasets that contain misinformation and show how system behavior emerges from the interaction between data characteristics and algorithmic design. Along similar lines, Fernández et al. (2021) study how recommendation choices affect content visibility and show that recommender mechanisms can unintentionally amplify misleading information.

Other work has focused more directly on collective dynamics. Bolzern et al. (2023) introduce a framework for multi-opinion Markovian agent networks, highlighting how opinions evolve and how influence is distributed unevenly across a population. Although not centered on AI-driven platforms, their formalism is relevant for modeling trust-related shifts at the population level. Likewise, Horta Ribeiro et al. (2023) show through agent-based simulation that recommender systems may either amplify or contain misinformation depending on configuration and context, while Pathak et al. (2023) examine how alternative recommendation strategies shape misinformation diffusion through user interaction patterns.

At a broader methodological level, Gribaudo et al. (2019) argue that complex social and behavioral systems often require multiple modeling perspectives, including network models, agent-based simulation, Petri nets, and Bayesian methods. Taken together, these studies show the value of simulation for understanding algorithmic influence and misinformation spread, but they leave comparatively underexplored the question of how trust-related outcomes depend on the joint timing of recommender escalation, moderation, and policy response within a unified formal framework.

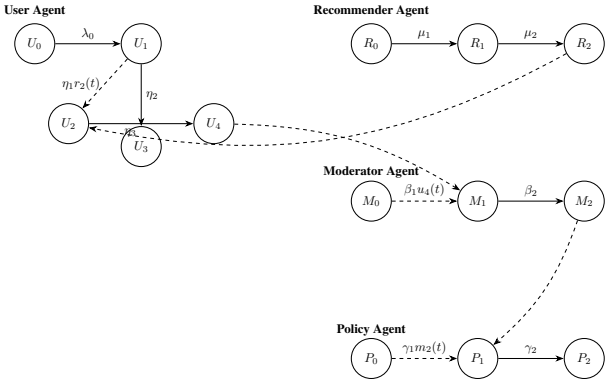


Figure 1: MAM for misinformation propagation in recommender systems. Dashed arrows denote state-dependent, cross-agent interactions.

3 Model

This section introduces the MAM used to study trust-related dynamics in online recommender systems. Trust is not modeled as a property of individual components or as a causal driver of behavior. Instead, it is evaluated at the system level as the outcome of interactions among users, recommender dynamics, moderation, and policy enforcement. The model is formulated in continuous time with state-dependent transition rates, yielding a closed mean-field system suitable for both simulation and structural analysis.

We consider an online platform in which exposure is mediated by a recommender optimized for engagement. Under sustained optimization pressure, the recommender may drift from a neutral state toward harmful amplification of sensational or misleading content. Moderation and platform-level policy enforcement respond only after harmful outcomes become visible, so their actions are reactive and delayed. The scenario is intentionally unfavorable and should be read as a stress test rather than as a general description of recommender systems.

The model in Figure 1 consists of four interacting agent classes: users, the recommender, moderators, and policy agents. Their states are listed in Table 1, while Table 2 reports the transition rates. Table 3 reports the baseline transition rates, delay values, and trust weights used in all simulations unless explicitly varied in the corresponding experiment. The corresponding mean-field equations are given in Table 4. Within each class, state variables represent conserved populations, ensuring well-posed dy-

Table 1: Agent classes and states in the misinformation MAM.

Agent Class	**States**
User Agent	U_0: Neutral (new user) U_1: Passive exposure U_2: Exposed to misinformation U_3: Resistant U_4: Believer
Recommender Agent	R_0: Neutral R_1: Engagement-optimized R_2: Harmful amplification
Moderator Agent	M_0: Passive monitoring M_1: Detection M_2: Content action
Policy Agent	P_0: Inactive P_1: Alerted P_2: Enforcement

Table 2: Continuous-time transition rates in the misinformation MAM.

Agent Class	**Transitions (rate)**
User	$U_0 \to U_1$: λ_0 $U_1 \to U_2$: $\eta_1 R_2(t)$ $U_1 \to U_3$: η_2 $U_2 \to U_4$: η_3
Recommender	$R_0 \to R_1$: μ_1 $R_1 \to R_2$: μ_2
Moderator	$M_0 \to M_1$: $\beta_1 U_4(t)$ $M_1 \to M_2$: β_2
Policy	$P_0 \to P_1$: $\gamma_1 M_2(t)$ $P_1 \to P_2$: γ_2

namics.[1]

Each agent class evolves along a one-directional state chain, with no endogenous backward transitions such as belief revision, de-amplification, or reversal of already activated institutional states. The model therefore represents escalation under delayed reactive governance in a conservative setting. Although corrective states can still grow and partially compensate harmful ones, full restoration becomes structurally difficult once harmful mass has accumulated.

3.1 Trust Index Definition

In this paper, trust is treated as a system-level evaluative observable. It is not modeled as an individual psychological attitude, nor as an internal state that feeds back into agent behavior. Instead, it is used as an explicit proxy to compare how different system trajectories unfold under the assumptions of the model.

[1]The code used to generate the results is publicly available at `https://github.com/EBarbierato/Modeling_Trust_in_AI_through_Markovian_Agent_Systems`.

Table 3: Baseline parameter values used in the misinformation simulations.

Parameter	Baseline value
λ_0	0.05
η_1	0.08
η_2	0.03
η_3	0.04
μ_1	0.06
μ_2	0.05
β_1	0.09
β_2	0.07
γ_1	0.08
γ_2	0.06
Moderation delay	0.0
Policy delay	0.0
w_b	1.0
w_a	0.6
w_m	0.3
w_p	0.4

Belief adoption is assumed to persist over the simulated horizon. This is a conservative choice: it isolates the effects of delayed governance in situations where misinformation outlasts institutional response times. The model does not exclude belief revision in real settings; it simply leaves it aside in order to focus on time-scale asymmetry.

Let $u_4(t)$, $r_2(t)$, $m_2(t)$, and $p_2(t)$ denote, respectively, belief prevalence, harmful amplification, moderation, and policy enforcement. These four quantities are used because they correspond to the harmful and corrective dimensions explicitly represented in the misinformation MAM. The terms $u_4(t)$ and $r_2(t)$ capture harmful outcomes at the user and recommender levels, while $m_2(t)$ and $p_2(t)$ represent the two corrective institutional layers.

The trust index is defined as

$$T(t) = \mathrm{clip}\Big(1 - \big(w_b u_4(t) + w_a r_2(t)\big) + \big(w_m m_2(t) + w_p p_2(t)\big), \; 0, 1\Big). \tag{1}$$

The weights control the relative contribution of each term. Under this construction, trust decreases when harmful belief and amplification increase, and increases when moderation and policy enforcement become active. Clipping keeps the index on the unit interval and avoids overinterpreting temporary overshoots.

The index should therefore be read as a diagnostic and comparative quantity: a model-specific baseline operationalization of trust, defined over harmful and corrective macro-states. The simulation results are interpreted relative to this operationalization and to the architecture of the model.

Table 4: Mean-field differential equations for the misinformation MAM.

User Agent Dynamics	
$u_0'(t) = -\lambda_0 u_0(t)$	Users leave the neutral entry state and begin content exposure.
$u_1'(t) = \lambda_0 u_0(t) - (\eta_1 r_2(t) + \eta_2) u_1(t)$	Passive exposure with competing misinformation and resistance.
$u_2'(t) = \eta_1 r_2(t) u_1(t) - \eta_3 u_2(t)$	Transition from exposure to belief.
$u_3'(t) = \eta_2 u_1(t)$	Development of resistance.
$u_4'(t) = \eta_3 u_2(t)$	Adoption of misinformation beliefs.
Recommender Agent Dynamics	
$r_0'(t) = -\mu_1 r_0(t)$	Drift toward engagement optimization.
$r_1'(t) = \mu_1 r_0(t) - \mu_2 r_1(t)$	Escalation toward amplification.
$r_2'(t) = \mu_2 r_1(t)$	Saturation in harmful amplification.
Moderator Agent Dynamics	
$m_0'(t) = -\beta_1 u_4(t) m_0(t)$	Activation triggered by belief prevalence.
$m_1'(t) = \beta_1 u_4(t) m_0(t) - \beta_2 m_1(t)$	Detection phase.
$m_2'(t) = \beta_2 m_1(t)$	Moderation actions deployed.
Policy Agent Dynamics	
$p_0'(t) = -\gamma_1 m_2(t) p_0(t)$	Policy activation.
$p_1'(t) = \gamma_1 m_2(t) p_0(t) - \gamma_2 p_1(t)$	Enforcement preparation.
$p_2'(t) = \gamma_2 p_1(t)$	Enforcement applied.

4 Results

This section presents simulation outcomes for the continuous-time MAM introduced in Section 3. We report a baseline configuration and ablation regimes meant to separate the structural contributions of moderation and policy enforcement. The focus stays at the population level, and we use the trust index to track how harmful and corrective system states balance over time.

Figure 2 reports the mean-field evolution of the user, recommender, moderation, and policy states under the baseline setting. The dynamics unfold as a cascade. The recommender moves from neutrality to engagement optimization and then into harmful amplification. As amplification grows, transitions from passive exposure into misinformation-related states accelerate, and belief adoption rises at the population scale. The trust curve in Figure 3 follows the same sequence. Trust starts high, then drops sharply once harmful amplification becomes prevalent, reaching a minimum of about 0.252. This drop occurs before moderation and policy enforcement fully activate, which is exactly where the adopted operationalization makes delay visible: corrective feedback exists, but it arrives late relative to the speed of harmful escalation. After institutional responses ramp up, the evaluated trust index rebounds, but it does not return to its initial level and instead settles around 0.553. Under persistent belief adoption, the trajectory becomes path-dependent: once a large belief mass has formed, later interventions

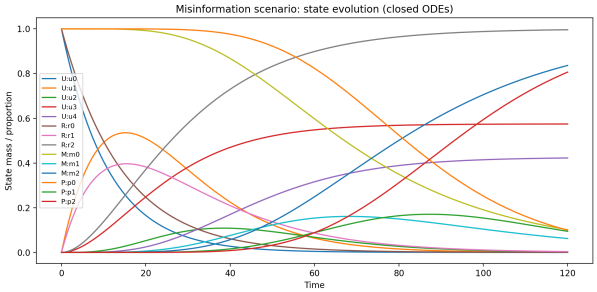


Figure 2: Mean-field state evolution under the baseline configuration. Harmful amplification precedes the activation of moderation and policy mechanisms, allowing misinformation belief to accumulate.

mainly slow further deterioration and enable only partial recovery.

Table 5 aggregates trust statistics for baseline and ablation settings. If moderation is disabled, the trust index declines without reversal and reaches zero within the simulated horizon. No recovery phase appears because the model then contains no effective corrective pathway that engages early enough to counter belief accumulation. If policy enforcement is removed while moderation remains active, the system shows a small rebound, but the final trust level is much lower (about 0.231). These ablations separate the functions of the two governance layers in the architecture studied here. Moderation prevents complete

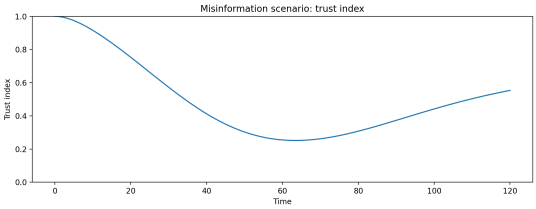


Figure 3: Trust index over time in the baseline regime. The index is a model-specific evaluative observable derived from harmful and corrective macro-states. Under the baseline configuration, it declines to approximately 0.252 before partially recovering to about 0.553.

collapse, but policy enforcement is what allows trust to stabilize at a non-trivial long-run level.

4.1 Sensitivity and Robustness Analysis

We performed a one-at-a-time sensitivity analysis by perturbing each transition rate around its baseline value while leaving all other parameters unchanged. Across the tested perturbations, the baseline qualitative pattern remains: when both moderation and policy enforcement are present, trust degrades and then partially recovers. In that sense, the regime is locally stable. The parameters associated with amplification and belief formation mainly change how quickly trust drops early on, whereas moderation and policy parameters have the strongest effect on the trust level reached by the end of the simulation horizon. Within the explored region, this suggests that trust resilience, as measured by the index, is shaped more by institutional feedback than by fine adjustments to the recommender dynamics.

4.2 Intervention–Timing Stress Test

We also stress-tested the model by introducing additional delays in the activation of moderation and policy enforcement, while keeping all other parameters fixed. In the current implementation, these delays are modeled as delayed onsets of the corresponding activation pathways: before the delay threshold is reached, the relevant activation flow is suppressed, while after the threshold the mechanism proceeds at its baseline rate. Figure 4 compares the resulting trust trajectories for the baseline, delayed moderation, and delayed policy settings. Within the explored delay range, the three trajectories remain very close to one another and the qualitative regime does not change. Delayed moderation produces the largest deviation from the baseline, but the effect remains limited.

Under this formulation, moderate response delays matter mainly as bounded perturbations around an already path-dependent trajectory. Once amplification and belief adoption have moved into a high-prevalence region, later activation of reactive interventions produces only small quantitative deviations in the trust curve. This result should be interpreted carefully, since in the present model,

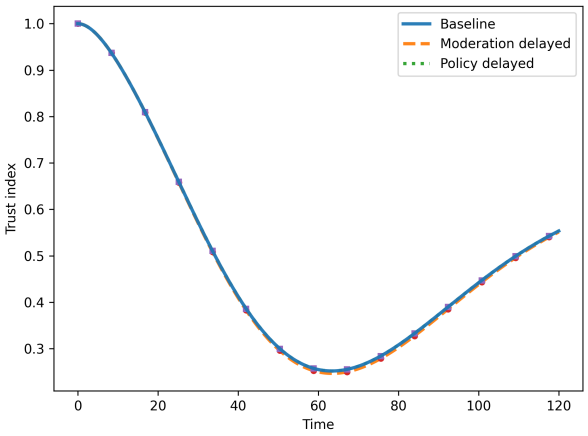


Figure 4: Trust trajectories under baseline conditions, delayed moderation, and delayed policy enforcement. Moderate delays introduce small deviations from the baseline, while trust dynamics remain largely path-dependent once amplification becomes dominant.

delay affects the onset of intervention, but not its post-activation strength or efficiency. As a result, the timing stress test supports a narrower conclusion about delayed activation.

4.3 Structural Properties of the Model

The mean-field system conserves mass within each agent class and preserves non-negativity of all state variables, so the dynamics remain well posed. Belief adoption is persistent over the simulated horizon, which encodes a conservative setting where belief consolidation outlasts the activation time of governance mechanisms. Combined with the explicit time-scale separation between fast recommendation dynamics and slower institutional response, this structure explains why the trust trajectories exhibit path dependence in the simulations, without implying a universal notion of hysteresis. The current formulation uses a feed-forward cascade architecture, which implies that each agent class evolves along a one-directional state chain, and the model does not include endogenous backward transitions. This design choice is analytically convenient because it isolates escalation and delayed intervention under persistent belief adoption, but it also constrains the class of trajectories the model can generate. In particular, the result that trust does not fully recover over the simulated horizon should be read as partly architecture-dependent: once harmful belief mass has accumulated, the absence of reversal pathways makes complete restoration structurally difficult. By contrast, the qualitative finding that corrective institutional layers play complementary roles, and that delayed reactive intervention allows harmful dynamics to consolidate before recovery mechanisms are active, is more robust within the present family of simulations. Adding belief revision, de-amplification, or other recovery transitions would be necessary to test whether incomplete recovery persists under less restrictive assump-

Table 5: Trust statistics under baseline and ablation regimes.

Scenario	Min Trust	Mean Trust	Final Trust
Baseline	0.252	0.490	0.553
No moderation	0.000	0.303	0.000
No policy enforcement	0.183	0.397	0.231

Table 6: Top parameters by sensitivity of the final trust index (one-at-a-time analysis).

Parameter	Impact on Final Trust
Policy enforcement rate (γ_2)	High
Policy activation rate (γ_1)	High
Moderation escalation rate (β_2)	Medium
Moderation activation rate (β_1)	Medium
Amplification rate (μ_2)	Low

tions.

5 Discussion

Under the adopted operationalization, the simulations frame trust as a system-level and path-dependent evaluation of how recommendation dynamics, belief formation, and institutional response unfold over time. Trust degradation does not arise from a single transition or parameter, but from the interaction and timing of multiple feedback processes. At the same time, these conclusions should be read carefully. The scenario is intentionally pessimistic: it assumes persistent belief adoption, one-directional state chains, and reactive governance that activates only after harmful dynamics become visible. The reported trajectories are therefore best interpreted as stress-test outcomes for an unfavorable socio-technical regime, not as a general characterization of trust dynamics across all AI-driven platforms.

The interpretation of the results also depends on two modeling choices. First, trust is represented through an explicit operational proxy built from the harmful and corrective macro-states of the model, rather than through a validated psychological or organizational measure. Second, the intervention–timing experiments model delay as deferred activation of corrective pathways, not as reduced intervention quality after activation. Within this setting, early amplification leaves lasting traces in the population state, so that later institutional response can produce only partial recovery.

Two conclusions are comparatively robust within the present model. Harmful amplification can outpace corrective layers when governance is reactive and delayed, and moderation and policy enforcement play complementary roles in limiting trust degradation. By contrast, the fact that trust does not return to its initial level depends more strongly on the irreversible state structure and the assumption of persistent belief adoption.

The ablation experiments clarify the functions of the two governance layers. Removing moderation eliminates the first effective corrective pathway and leads to sustained trust degradation. Removing policy enforcement still allows some recovery, but at a much lower trust level. This indicates that moderation acts locally by constraining exposure, whereas policy enforcement operates at a broader systemic level. Trust resilience therefore depends on the presence of both.

The sensitivity analysis shows that amplification and belief-formation parameters mainly shape the speed of early decline, whereas moderation and policy parameters most strongly influence the trust level reached at the end of the simulation horizon. The timing stress test further shows that, within the explored range, delayed activation produces only limited deviations from baseline. Once harmful dynamics become dominant, later intervention can slow deterioration and enable partial recovery, but not restore the initial trust level.

6 Conclusion

Under the adopted operationalization and within the conservative stress-test regime studied here, trust declines as harmful amplification takes hold and recovers only partially even when corrective mechanisms are active.

The simulations show that moderation and policy enforcement play distinct and complementary roles. Moderation limits exposure and prevents complete collapse, while policy escalation is needed to stabilize behavior over longer time horizons. In this setting, trust-related breakdowns arise mainly from structural features of the system, especially the timing and coordination of oversight mechanisms, rather than from isolated algorithmic choices.

More generally, the results indicate that, in a reactive governance architecture, improving trust outcomes cannot be achieved through algorithmic refinement or content filtering alone. What matters most is whether intervention pathways activate early enough and in a coordinated way before harmful dynamics become dominant. Within the range explored here, accelerating reactive mechanisms after amplification and belief adoption are already established yields only limited gains.

These conclusions should be read relative to the present assumptions and to the deliberately pessimistic character of the scenario. At the architectural level, the simulations indicate the importance of delayed reactive governance and the complementary roles of moderation and policy enforcement. At the trajectory level, however, incomplete

recovery depends more strongly on the conservative, one-directional formulation adopted here. Future work could extend the framework through heterogeneous user populations, feedback loops in which trust affects engagement or intervention intensity, empirical grounding of $T(t)$, and comparison with alternative modeling formalisms.

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